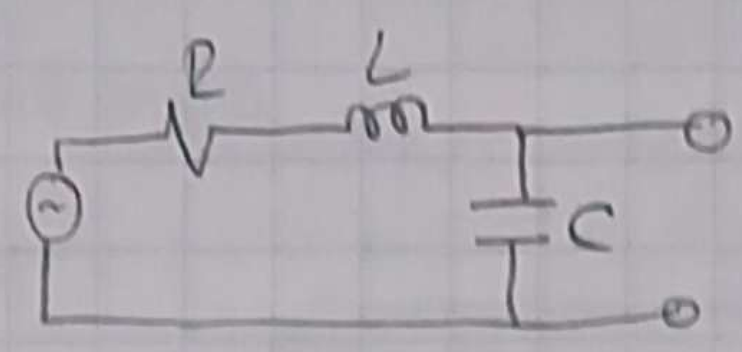
$V_i(t)$ 

$V_c(t)$

EDO
↳ Ecuación diferencial ordinaria

Tomando en cuenta que es una ecuación que relaciona una señal $V_c(t)$ con sus derivadas en el tiempo

$R \to V_R(t) = R\,i(t)$
$L \to V_L(t) = L\, di/dt\,(t)$
$C \to i(t) = C\, dV_c/dt\,(t)$ -> será la salida => $V_c(t) = V_c(t)$

se aplica ley de voltajes de Kirchhoff
↳ $V_L + V_R + V_c = V_i$ } -> entrada

Sustitución en función V_c

$$LC\,\frac{d^2 V_c(t)}{dt^2} + RC\,\frac{dV_c(t)}{dt} + V_c(t) = V_i(t)$$

↳
$$\frac{V_i(t)}{LC} = RC\,\frac{dV_c(t)}{dt} + LC\,\frac{\frac{d^2 V_c(t)}{dt}}{LC} + \frac{V_c(t)}{LC}$$

$$\frac{1}{LC}\,V_i(t) = V_c''(t) + \frac{R}{L}\,V_c'(t) + \frac{1}{LC}\,V_c(t)$$

Función de transferencia $H(\omega) = \frac{Y(\omega)}{X(\omega)}$

$$H(s) = \frac{Y(s)}{X(s)}$$ -> salida / -> entrada

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\frac{1}{LC}}{V_c''(t) + \frac{R}{L}\,V_c'(t) + \frac{1}{LC}\,V_c(t)}$$

- Diagrama de bode

↳ sus gráficas son de función de transferencia:

1. Magnitud vs frecuencia
2. Fase vs frecuencia

Magnitud en dB

$$dB = 20 \log_{10} |H(jw)|$$

fase $\to H(jw) \to \underline{\angle H(jw)}$

- Función de transferencia con Laplace y polos y ceros

Condiciones iniciales $= 0$

$$\mathcal{L}\{V_c(t)\} = V_c(s)$$

$$\mathcal{L}\left\{\frac{d\,V_c(t)}{dt}\right\} = sV_c(s)$$

$$\mathcal{L}\left\{\frac{d^2}{dt^2} V_c(t)\right\} = s^2 V_c(s)$$

$$\mathcal{L}\{V_i(t)\} = V_i(s)$$

$R = 1000\,\Omega$
$L = 0.18\,H$
$C = 120 \times 10^{-6}\,F$

$$LC\frac{d^2}{dt^2}V_c(t) + RC\frac{d}{dt}V_c(t) + V_c(t) = V_i(t)$$

$$LC\,s^2\,V_c(s) + RC\,sV_c(s) + V_c(s) = V_i(s)$$

$$V_c(s)\,[LCs^2 + RCs + 1) = V_i(s)$$

$$H(s) = \frac{V_c(s)}{V_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

②

Reemplazando

↳ $H(s) = \dfrac{1}{(0.18)(120\cdot10^{-6})s^2 + (1000)(120\times10^{-6})s+1} = \dfrac{1}{2.16\times10^{-5}s^2 + 0.12s+1}$

Ceros: raices en el numerador → No hay

Polos: Raices en el denominador: $\overbrace{2.16\times10^{-5}}^{a}s^2 + \overbrace{0.125}^{b} + \overbrace{1}^{c}$

$$s = \frac{-0.12 \pm \sqrt{0.12^2 - 4(2.16\times10^{-5})(1)}}{2(2.16\times10^{-5})}$$

$s_1 \simeq -8.35$ $\quad s_2 \approx -5547.21$

· Tenemos entrada el cual es respuesta al impulso $(h(t))$

$x(t) = \delta(t)$

$\mathcal{L}\{\delta(t)\} = X(s) = 1$ → Despejando la salida $Y(s) = H(s)\cdot X(s)$

$Y(s)\ H(s) \cdot 1 = H(s)$

$h(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\{H(s)\}$

$$H(s) = \frac{1}{2.16\times10^{-5}s^2 + 0.12s+1}$$

$s_1 = -8.35$ rad/s $\quad s_2 = -5547.21$ rad/s

$$H(s) = \frac{1}{as^2+bs+c} = \frac{1}{(s-s_1)(s-s_2)} = \frac{A}{(s-s_1)} + \frac{B}{s-s_2}$$

$$= \frac{A}{s+8.35} + \frac{B}{s+5547.21}$$

$$K = \frac{1}{2.16\times10^{-5}} \simeq 46296.30$$

$$\frac{K}{(s-s_1)(s-s_2)} = \frac{A}{s-s_1} + \frac{B}{s-s_2}$$

Para A :

↳ $K = A(s-s_2) + B(s-s_1)$

$K = A(s-s_2)$

$$A = \frac{K}{(s-s_2)} = \frac{46296.30}{-8.35-(-5547.21)} \simeq \boxed{8.35}$$

Para B :

$K = A(s-s_2) + B(s-s_1)$

$K = B(s-s_1)$

$$B = \frac{K}{(s-s_1)} = \frac{46296.30}{-5547.26-(-8.35)} \simeq -8.35$$

$$\frac{1}{s+a} \rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}\, a(t)$$

$$H(s) = \frac{A}{s+8.35} + \frac{B}{s+5547.21} = A e^{-8.35t}\, a(t) + B e^{-5547.21t}\, a(t)$$

$A = 8.35$ $\quad$ $B = -8.35$

$$\frac{K}{s(s-s_1)(s-s_2)} = \frac{A}{s} + \frac{B}{s-s_1} + \frac{C}{s-s_2}$$

↳ Multiplicamos esto por el numerador para cancelar

Para A $(s=0)$ = $A = \frac{K}{(-s_1)(-s_2)} = \frac{46296.3}{(-8.35)(-5547.21)} = 1$

para B $(s=s_1)$ = $B = \frac{K}{s(s-s_2)} = \frac{46296.3}{-8.35(-8.35-(-5547.21))} \approx -1.0015$

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Para $C(s-s_2) = C = \frac{K}{s_2(s_2-s_1)} = \frac{46296.3}{-5547.21(-5547.21-(-8.351))} \approx$

$$\boxed{= 0.00151}$$

$$Y(s) = \frac{A}{s} + \frac{B}{s-s_1} + \frac{C}{s-s_2} = \frac{1}{s} - \frac{1.0015}{s-s_1} + \frac{0.00151}{s-s_2}$$

$$y(t) = A \cdot u(t) + B e^{s_1 t} u(t) + C e^{s_2 t} u(t)$$

$$y(t) \approx [1 - 1.0015 e^{s_1 t} + 0.00151 e^{s_2 t}] u(t)$$

$$y(t) = [1 - 1.0015 e^{-8.35t} + 0.0015 + e^{-5547.21} t] u(t)$$